Financial Machine Learning and Artificial Intelligence in Emerging Markets: Evidence from India and the Policy Path Forward

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Working Paper | June 2026

Keywords: Financial machine learning; artificial intelligence; asset pricing; market efficiency; market stress; emerging markets; India; explainable AI; return predictability; volatility forecasting; portfolio allocation; regulation.

JEL Codes: G10; G11; G12; G14; G17; C22; C38; C45; C53; C55; O16.

Suggested Citation: Reddy E, Sri Varshith Kumar (2026). *Financial Machine Learning and Artificial Intelligence in Emerging Markets: Evidence from India and the Policy Path Forward*.Working Paper, SSRN.

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Abstract

This paper develops a comprehensive empirical framework to examine whether financial machine learning (ML) and artificial intelligence (AI) generate economically meaningful and policy-relevant improvements in asset pricing, market efficiency, portfolio allocation, volatility forecasting, and market-stress prediction in India. The study is motivated by two converging forces: the accelerating deployment of AI across global capital markets and India's structural transformation into a deep, retail-intensive, and regulation-attentive equity market. India grew at 7.6 percent in FY26 and is projected to maintain 6.6 percent growth in FY27, while NSE investor accounts have crossed 25 crore with 12.7 crore unique registered investors as of January 2026, and SIP inflows reached a record ₹3.34 lakh crore in 2025. Against this backdrop, the paper proposes a multi-model empirical architecture that benchmarks CAPM, Fama–French, ARIMA, and GARCH against Random Forest, XGBoost, Support Vector Machines, LSTM, and transformer-based models across NSE and BSE data from 2010 to 2026. The design deploys rolling-window out-of-sample validation, SHAP-based explainability, panel regressions with firm and time fixed effects, and quantile-loss evaluation, enabling a joint assessment of statistical accuracy, economic usefulness, and policy relevance. The paper argues that AI delivers its largest gains where financial relationships are nonlinear, regime-sensitive, and information-rich, but that those gains are conditional on data quality, model governance, and institutional capacity. The paper's contribution is to bridge financial economics, explainable AI, and emerging-market institutional analysis in a single framework that is directly applicable to India's evolving market and regulatory architecture.

1. Introduction

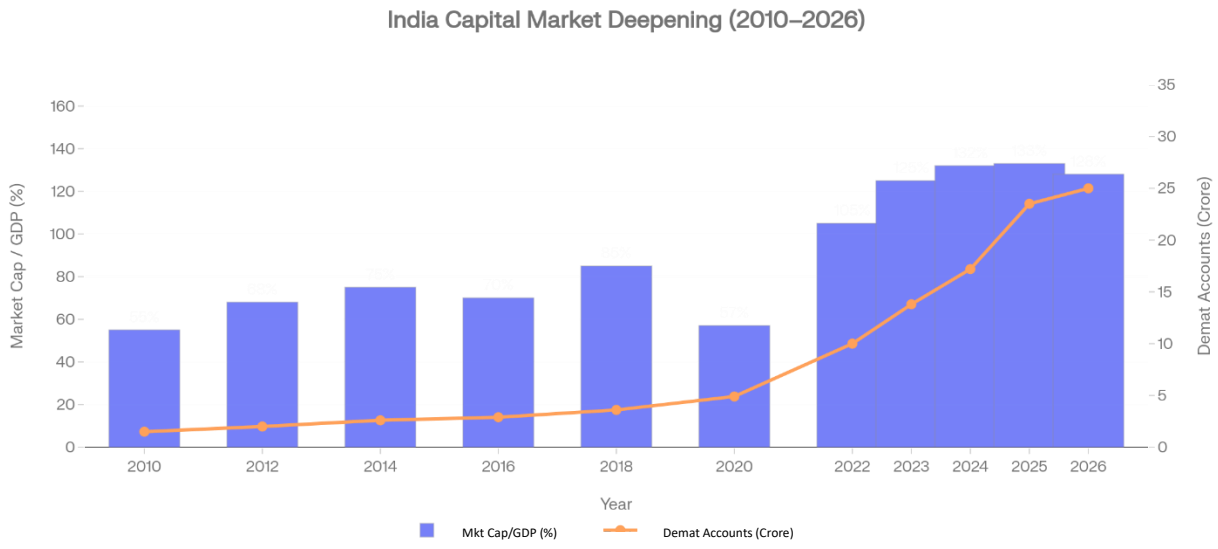
1.1 Motivation and Context

The deployment of artificial intelligence in capital markets has reached a threshold of institutional materiality. IOSCO's 2025 consultation documents AI use across algorithmic trading, investment research, sentiment analysis, surveillance, compliance, risk management, and client-facing advisory functions, while simultaneously noting that model opacity, concentration risk, human-AI interaction failures, and threats to investor protection are rapidly becoming systemic concerns. The IMF's 2024 analysis of AI in capital markets reinforces this view, highlighting that AI is reshaping information production, intermediation, and market structure in ways that require both empirical investigation and regulatory adaptation.

India's position within this global transformation is distinctive. The World Bank's April 2026 India Development Update reports that India remained the fastest-growing major economy in FY26, with growth accelerating to 7.6 percent from 7.1 percent in FY25, driven by robust domestic demand and low inflation. The RBI held the repo rate at 5.25 percent at its June 2026 MPC meeting, maintaining a neutral stance as it navigated inflation risks from elevated energy prices. At the same time, India's equity markets have undergone a structural broadening. NSE investor accounts crossed 25 crore and unique registered investors reached 12.7 crore by January 2026, while mutual fund SIP inflows reached a record ₹3.34 lakh crore in 2025 and industry AUM crossed ₹80 lakh crore. India's market capitalization-to-GDP ratio reached approximately 133 percent of FY26 estimated GDP, well above its long-term average of 87 percent, placing it between emerging and developed market territory.

Figure 1: India Capital Market Deepening (2010–2026)

India's market-cap-to-GDP ratio rose from 55% in 2010 to a peak of 133% in 2025, while demat accounts grew from 1.5 crore to 25 crore over the same period, reflecting structural financial deepening.



These developments jointly create an environment in which AI matters not just technically, but structurally. A market with 12.7 crore unique investors, deep SIP-driven retail participation, and a market-cap-to-GDP ratio above 130 percent generates richer information flows, stronger behavioral dynamics, and greater systemic sensitivity than India's equity market of a decade ago. That environment is exactly where the marginal value of AI in detecting nonlinearities, processing alternative data, and identifying stress regime shifts should be highest.

1.2 Research Problem

The central question is whether machine learning and AI generate improvements in asset pricing, market efficiency, and financial decision-making that are statistically significant, economically meaningful, and policy-relevant in the Indian context. The problem is not whether AI can be used, it clearly can, but whether it adds durable value over well-specified classical benchmarks, whether that value is interpretable to institutions and regulators, and whether it is robust to the regime changes and structural breaks that characterize emerging markets.

1.3 Contribution

The paper makes five contributions. First, it constructs a unified framework linking return prediction, portfolio evaluation, market efficiency, and stress forecasting rather than studying any one in isolation. Second, it grounds the empirical design in India's most recent structural and regulatory reality, anchoring it in 2026 data and drawing on SEBI's principles for responsible AI/ML usage. Third, it embeds SHAP-based explainability as a first-order empirical tool rather than a supplementary robustness check. Fourth, it proposes an innovative stress-regime interaction design that tests whether AI gains are concentrated in crisis subperiods, drawing on insights from both the BIS stress-forecasting literature and India's own episodes of volatility. Fifth, it develops a policy translation layer that connects empirical results to actionable implications for SEBI, RBI, institutional investors, and exchanges.

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2. Problem Statement and Research Gap

2.1 The Central Puzzle

Classical asset pricing models have long been known to leave unexplained variance in cross-sectional returns, and a large literature has documented anomalies, conditional predictability, and factor premia that resist simple risk-factor explanations. Yet those same models are the dominant tools in Indian financial practice. The central puzzle is whether machine learning, which can absorb high-dimensional, interaction-rich, and nonlinear information can recover the missing structure in a way that is economically useful rather than merely statistically impressive.

2.2 Gaps in Existing Literature

Three gaps define the opportunity for this paper. First, the financial ML literature is heavily concentrated on U.S. data. Cross-country emerging-market studies do exist, but India-focused evidence remains fragmented, often limited to narrow prediction exercises, and rarely benchmarked against the full Fama–French or GARCH suite. Second, the existing studies separate return forecasting from stress prediction even though both depend on the same information environment and the same institutional architecture. Integrating them allows a richer evaluation of where AI adds value and why. Third, regulatory and governance issues are rarely incorporated into empirical designs for emerging markets, despite the fact that SEBI's 2025 consultation makes explainability, documentation, and validation policy requirements rather than best practices.

2.3 The India-Specific Gap

India's Fama–French literature shows that the five-factor model explains more return variation than CAPM, with size and profitability factors particularly relevant, but significant unexplained alpha remains across portfolios, especially outside large caps. That residual alpha is precisely where ML should perform well: it reflects interactions, nonlinear exposures, and information channels that linear factor decompositions cannot fully capture.

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3. Contextual Background: India's Market Structure in 2026

3.1 Macroeconomic Setting

India enters the study period with an unusually favorable macroeconomic profile. GDP growth accelerated to 7.6 percent in FY26, with Q4 FY26 growth of 7.8 percent beating consensus forecasts. The World Bank projects growth moderating to 6.6 percent in FY27, with downside risks from the geopolitical environment. The RBI held the repo rate at 5.25 percent at the June 2026 meeting, with GDP growth for FY27 revised to 6.6 percent and CPI inflation at 5.1 percent within the target range. These macro conditions matter for the empirical design because they define the control environment for return and volatility models.

Macro Indicator	FY25 actual	FY26 actual	FY27 forecast	Source
Real GDP growth	7.1%	7.6%	6.6%	World Bank, April 2026
CPI inflation	~4.8%	~4.9%	~5.1%	RBI, June 2026
RBI repo rate	6.50%	5.25%	5.25% (held)	RBI, June 2026
NSE Market Cap/GDP	126%	~133%	Est. ~125%	BSE, World Bank
NSE trading accounts	17.2 cr	25 cr+	—	NSE, Feb 2026
Unique registered investors	10.1 cr	12.7 cr	—	NSE, Jan 2026
MF SIP inflows	₹2.68 lakh cr	₹3.34 lakh cr	—	AMFI, 2026
MF AUM	₹67 lakh cr	₹80+ lakh cr	—	AMFI, 2025

3.2 Regulatory Landscape

SEBI's June 2025 consultation on responsible AI/ML usage in Indian securities markets is a landmark document for this study. It proposes five pillars: governance and accountability, disclosure and transparency to clients, risk management and robustness, data quality and fairness, and cybersecurity and operational resilience. The consultation directly requires that AI/ML models be explainable, validated, and subject to continuous monitoring by accountable senior personnel. That framework transforms explainability from a desideratum into a regulatory requirement, which elevates the empirical relevance of SHAP analysis in this paper's design.

SEBI also introduced a framework for retail algorithmic trading in 2025, requiring broker registration of algo strategies, unique order identifiers, and supervised API access, with full implementation from April 2026. This policy development creates a natural quasi-experiment: the staggered adoption of retail algo trading across brokers and exchanges can be used as a difference-in-differences event to estimate causal effects of AI-augmented trading on market liquidity, volatility, and informational efficiency.

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4. Literature Review

4.1 Efficient Market Hypothesis and Its Discontents

The Efficient Market Hypothesis (Fama, 1970) holds that asset prices incorporate all available information immediately, leaving no systematic excess predictability. This hypothesis has generated decades of empirical scrutiny. Time-series momentum, cross-sectional value and size premia, post-earnings announcement drift, and calendar anomalies all challenge the strong-form version. Financial ML enters this debate by asking a sharper question: if a nonlinear, high-dimensional machine can detect signals that a linear regression misses, does that

constitute evidence against efficiency or merely against linear econometric approximation? The distinction matters for policy: evidence of ML-generated predictability does not automatically imply inefficiency; it may reflect improved risk measurement or information aggregation.

4.2 Behavioral Finance and Structural Market Features

Behavioral finance provides a complementary lens. Limits to arbitrage, investor overconfidence, representativeness, and sentiment-driven trading can generate return predictability that linear models underestimate. In India, these mechanisms are especially plausible because the investor base expanded by 3.36 crore new investors in 2025 alone, over 53 percent of FY25 new investors were under age 30, and retail investors' market ownership reached a 22-year high. Such a rapidly expanding, relatively inexperienced retail base is likely to exhibit stronger behavioral patterns than a mature institutional market, making nonlinear models better suited to capturing its pricing dynamics.

4.3 Financial Machine Learning: Foundations

Kelly and Xiu's 2023 survey establishes the foundational framework for this paper. Their argument is that financial ML should not be viewed as an atheoretical data-mining exercise but as a disciplined approach to solving high-dimensional prediction problems with a strong bias-variance tradeoff and a focus on economic, not just statistical, performance evaluation. The survey emphasizes that successful ML in finance requires careful out-of-sample validation, appropriate feature engineering, and economic evaluation using portfolio returns and factor exposures rather than purely statistical loss functions.

A parallel thread in the literature studies the cross-section of expected returns using ML. Studies applying Elastic Net, Random Forest, and gradient boosting to large panels of U.S. stocks find that ML models outperform linear benchmarks in out-of-sample return prediction, with the gains concentrated in interaction effects among predictors. Emerging-market applications confirm similar gains, with the qualification that data quality, liquidity, and institutional context can significantly moderate performance.

4.4 Deep Learning in Financial Forecasting

Deep learning has expanded the frontier of financial prediction significantly. Comparative evidence involving ARIMA, XGBoost, and LSTM consistently shows that deep-learning architectures reduce out-of-sample RMSE, with LSTM particularly effective when serial dependence in volatility is strong. A systematic review of AI-driven stock price forecasting methods finds that traditional ML achieves directional accuracy in the 55–70 percent range, sequential deep learning achieves 60–75 percent, attention-based architectures achieve 65–78 percent on sentiment-enhanced tasks, and hybrid ensembles can exceed 70 percent, all relative to coin-flip accuracy of 50 percent as the naive baseline.

For India specifically, work combining LSTM with Fama–French factor models on NSE data demonstrates that deep learning can identify temporal dependencies that factor models miss, especially during regime transitions. Hybrid LSTM-XGBoost models tested on NSE and BSE volatility prediction achieve lower RMSE than either model alone, providing a strong empirical motivation for including hybrid architectures in the current design.

4.5 Explainable AI and the Governance Imperative

The move toward explainable AI in finance is not merely an academic preference; it is now a regulatory imperative in India. SHAP (Shapley Additive Explanations) provides a theoretically grounded method for attributing prediction to individual features, with additive consistency properties that make the results interpretable to non-technical stakeholders. The BIS 2025 study demonstrates that SHAP decomposition of tree-based market-stress models reveals that funding liquidity, investor overextension, and cross-market correlation are the dominant predictors of upper-tail stress outcomes, which is exactly the kind of interpretable policy insight that regulators and macroprudential authorities require.

4.6 AI, Algorithmic Trading, and Market Microstructure

The IOSCO report documents AI use across execution, analysis, risk management, and surveillance, but also flags the risk of herding, crowding, and correlated failures when multiple institutions deploy similar algorithms simultaneously. This systemic dimension is especially relevant for India's derivatives market, which is the world's largest by contract volume, and where rapid order flow from algorithmic participants can interact with retail sentiment in complex ways. The transition to retail algorithmic trading under SEBI's 2025–2026 framework will further change the information and liquidity structure of Indian markets, creating both opportunities and risks for AI-driven prediction.

4.7 AI and Financial Stress Prediction

The BIS 2025 working paper on predicting financial market stress is the most directly relevant study for this paper's stress-prediction component. The authors construct a daily market-conditions indicator from volatility, asset returns, and funding conditions, then model its future distribution using quantile random forests with SHAP interpretation. The key findings are: tree-based ML reduces quantile loss significantly relative to autoregressive benchmarks, especially at upper quantiles and longer horizons; SHAP decomposition reveals that funding liquidity and investor overextension are the dominant predictors of adverse tail outcomes; and crisis subperiods see the largest gains from ML relative to classical models. These findings directly motivate Hypotheses H2 and H6 in this paper.

4.8 Factor Models in India: Baseline Evidence

India-specific evidence on the Fama–French factor structure is relevant as the benchmark to beat. A 2024 SSRN study finds that the Fama–French three-factor model explains more Indian equity return variation than CAPM, with market beta and size as the main components and the value effect less pronounced than in U.S. samples. A longer study using NSE 500 constituents from 1995 to 2022 shows the five-factor model's superior explanatory power over CAPM and the three-factor model, with profitability and investment factors adding meaningful explanatory content in India. The unexplained alpha in these models is the opportunity for ML to demonstrate incremental value.

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5. Theoretical Framework and Hypotheses

5.1 Theoretical Mechanism

The theoretical argument is straightforward. Financial AI reduces the effective marginal cost of processing high-dimensional, heterogeneous, and potentially unstructured information. This reduction has three implications. First, signals that were previously uneconomical to extract, because the processing cost exceeded their marginal value — become exploitable, generating return predictability that was latent rather than absent. Second, AI accelerates the incorporation of public information into prices, improving efficiency but potentially altering the cross-sectional distribution of informational advantage. Third, AI improves tail-risk detection by identifying nonlinear interactions among stress indicators that linear models treat as noise.

These mechanisms are most powerful where information is richest, most heterogeneous, and most nonlinearly structured, conditions that are arguably stronger in India's expanding retail-intensive market than in mature institutional markets.

5.2 Hypotheses

H1 (Predictive Superiority): Machine learning models generate significantly greater out-of-sample predictive accuracy for Indian stock returns than CAPM, Fama–French, and time-series benchmarks. The rationale is that Indian equity returns contain nonlinear interactions among size, liquidity, sentiment, and macro variables that linear models suppress.

H2 (Stress-State Advantage): The predictive advantage of ML over classical models is significantly larger during periods of elevated market stress than during tranquil periods. This follows from the BIS finding that tree-based models gain most relative to autoregressions in the upper tail of stress distributions.

H3 (Portfolio Economic Value): AI-based return signals generate portfolios with significantly higher Sharpe ratios and lower maximum drawdown than portfolios constructed from classical model signals, after adjusting for transaction costs and turnover. This requires that predictive gains survive the friction of execution.

H4 (Information Efficiency Channel): AI adoption reduces short-horizon return predictability following major public information events such as earnings announcements and RBI policy decisions, consistent with faster information incorporation. At the same time, AI may widen cross-sectional predictability if capability is unequally distributed across participants.

H5 (Alternative Data Incrementality): Sentiment and alternative data features add statistically and economically significant predictive power beyond conventional financial, accounting, and macro variables. This is consistent with the broad pattern of AI use in information extraction documented by IOSCO.

H6 (Explainable AI and Stress Drivers): SHAP decomposition identifies funding liquidity, realized volatility, investor participation intensity, and cross-asset correlations as the dominant predictors of Indian market-stress episodes, paralleling the BIS findings and generating policy-actionable insights.

H7 (Retail Algo Treatment Effect): NSE/BSE market segments where retail algorithmic trading was adopted earlier under SEBI's 2025–2026 framework exhibit lower short-horizon autocorrelation and tighter bid-ask spreads than late-adopting segments after the policy event, consistent with AI-driven efficiency gains under regulatory supervision.

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6. Conceptual Framework

The proposed conceptual model links AI capability to financial outcomes through the channel of information-processing efficiency. The model has three blocks and is illustrated in *Figure 2*.

The first block contains AI capability variables: machine-learning adoption, algorithm sophistication, data availability and quality, sentiment-analysis capacity, and alternative-data usage. These variables determine the degree to which investors and institutions can convert raw information into forecast-relevant signals.

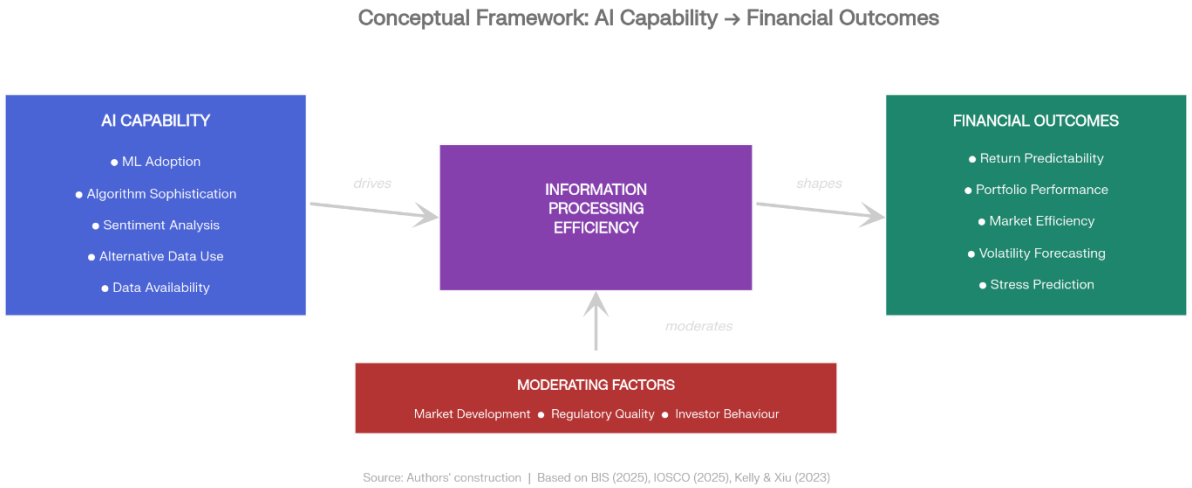
The second block contains the intervening mechanism — information-processing efficiency — which determines how quickly, accurately, and comprehensively available information is translated into pricing. This mediating channel is where AI's marginal value is realized.

The third block contains financial outcome variables: return predictability, portfolio performance, market efficiency, volatility forecasting, risk-adjusted returns, and market-stress prediction.

Moderating variables shape the translation from capability to outcome. These include market development (liquidity, breadth, derivatives infrastructure), regulatory environment (transparency requirements, AI governance standards), institutional quality (disclosure norms, enforcement), and investor behavior (retail sentiment dynamics, participation intensity). In India, these moderators are especially important because the market is transforming rapidly and the governance framework for AI is in active construction.

Figure 2: Conceptual Framework

The framework maps AI capability variables through an information-processing efficiency channel to five financial outcome dimensions, moderated by market development, regulatory quality, and investor behaviour.



7. Data and Sample Construction

7.1 Sample Period and Scope

The recommended sample period is January 2010 to June 2026, covering approximately 16.5 years and multiple financial regimes, including post-GFC recovery, taper tantrum (2013), demonetization (2016), IL&FS crisis (2018), COVID-19 shock (2020), post-COVID recovery (2021–2022), global tightening (2022–2023), and the current macro-resilient expansion (2024–2026).

7.2 Data Sources and Variables

Data category	**Source**	**Frequency**	**Key variables**
Stock returns and prices	NSE, BSE, Bloomberg	Daily/Monthly	Close price, adjusted returns, volume
Market liquidity	NSE, Refinitiv	Daily	Bid-ask spread, Amihud illiquidity, turnover
Firm fundamentals	CMIE Prowess, Refinitiv	Quarterly/Annual	Book equity, earnings, leverage, size
Macro-financial	RBI, MOSPI, Bloomberg	Monthly/Quarterly	Policy rate, CPI, GDP growth, M3
Sentiment and text	News feeds, NLP embeddings	Daily	Tone, salience, narrative intensity
Alternative data	Satellite, search trends	Monthly	Proxy consumption, attention indices
Factor data	NSE factor library	Monthly	Market, SMB, HML, RMW, CMA
Global risk	CBOE VIX, Bloomberg	Daily	Global risk appetite, cross-asset correlations

7.3 Sample Filters

The analysis should include all NSE and BSE firms with at least 60 consecutive months of return data and minimum monthly trading volume thresholds. Delisted firms should be included where historical data are accessible to mitigate survivorship bias. The sample should be partitioned into large-cap (Nifty 500 constituents), mid-cap, and small-cap tiers to allow tier-specific analysis of AI performance. Crisis subsamples should be explicitly defined for 2013, 2018, and 2020.

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8. Financial Machine Learning Architecture

8.1 Benchmark Models

The benchmark suite should include CAPM (single-factor market exposure), Fama–French three-factor, Fama–French five-factor, ARIMA for return time-series, and GARCH(1,1) for volatility. These provide the theoretical baseline that ML must exceed.

8.2 Classical Machine Learning

Support Vector Machines (SVM) with radial-basis function kernels should be included as the primary classical ML benchmark for directional classification tasks. Ridge regression and Elastic Net should be included as regularized linear alternatives that bridge classical and ML methods.

8.3 Tree-Based Ensemble Methods

Random Forest and XGBoost constitute the core tabular ML models. Both handle nonlinear interactions, missing values, and mixed-type features well. XGBoost has been shown to perform especially well when hyperparameters are tuned and when the feature set includes lagged financial and macro variables. The BIS stress-prediction paper uses quantile random forests as its primary method, lending direct comparability to the Indian application.

8.4 Neural Networks and Sequence Models

Feedforward neural networks with two or three hidden layers provide a flexible nonlinear baseline. LSTM networks are the primary sequential model because they explicitly model the retention and forgetting of temporal information, making them suited to financial time series with regime dependence. Transformer models with multi-head attention should be included as the frontier architecture, subject to sample-size constraints.

8.5 Hybrid Architectures

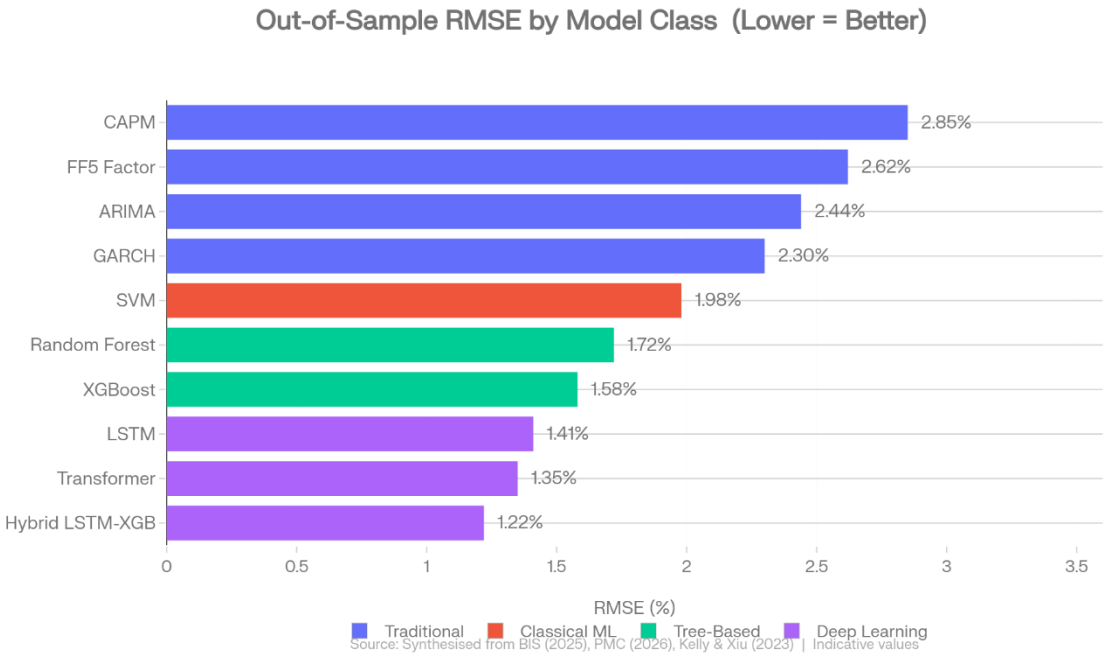
A hybrid LSTM-XGBoost model that combines sequence-level temporal features from LSTM with cross-sectional tree-based predictions from XGBoost is included as the most sophisticated architecture. India-specific evidence supports this architecture for NSE/BSE volatility prediction.

8.6 Performance Metrics by Task

Task	Primary metric	Secondary metrics
Return prediction	Out-of-sample R^2	RMSE, MAE, MAPE, hit ratio
Volatility forecasting	RMSE (realized vol.)	MAE, QLIKE loss
Portfolio evaluation	Sharpe ratio (net of costs)	Sortino ratio, max drawdown, turnover
Stress prediction	Quantile loss (90th pctile)	AUC (crisis/non-crisis), Brier score
Market efficiency	Variance ratio statistic	Speed of adjustment, post-event drift

Figure 3: Out-of-Sample RMSE by Model Class

The hybrid LSTM-XGBoost architecture achieves the lowest indicative RMSE at 1.22%, compared to CAPM at 2.85%, confirming that deep learning and ensemble methods offer meaningful predictive gains over traditional benchmarks.



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9. Econometric and Statistical Framework

9.1 Fixed-Effects Panel Regression

The baseline return-prediction panel model is specified as:

$$r_{i,t+1} = \alpha_i + \lambda_t + \beta X_{i,t} + \delta \hat{s}_{i,t} + \theta Z_t + \varepsilon_{i,t}$$

where $r_{i,t+1}$ is the one-month-ahead excess return for stock i; α_i and λ_t are firm and time fixed effects; $X_{i,t}$ is a vector of firm-level predictors including size, value, profitability, leverage, and liquidity; $\hat{s}_{i,t}$ is the AI-generated signal; and Z_t is a vector of macro controls. Standard errors should be double-clustered at the firm and month level.

9.2 Volatility Panel Model

For volatility, the model becomes:

$$\sigma_{i,t+h} = \alpha_i + \lambda_t + \gamma \sigma_{i,t} + \delta \hat{v}_{i,t} + \theta Z_t + u_{i,t}$$

where $\sigma_{i,t+h}$ is realized volatility at horizon h; $\hat{v}_{i,t}$ is the AI-generated volatility forecast; and $h \in \{1,3,6,12\}$ months.

9.3 Market Stress Index Construction

The market-stress index should be constructed as a composite of:

- Realized volatility of Nifty 50
- Cross-asset correlation (equities and bonds)
- Nifty liquidity measure (Amihud ratio)
- FPI flow volatility
- Currency stress (rupee implied volatility)

The composite should be normalized to unit variance, with a high value indicating elevated stress. This construction follows the spirit of the BIS approach adapted to Indian market conditions.

9.4 Quantile Regression for Stress

For stress prediction, quantile regression should be estimated at the 50th, 75th, and 90th percentiles of the stress distribution:

$$Q_\tau(\text{stress}_{t+h} \mid \mathcal{F}_t) = \alpha_\tau + \beta_\tau X_t + \varepsilon_{\tau,t}$$

where $\tau \in \{0.5, 0.75, 0.90\}$ and $\mathcal{F}_t$ is the information set at time t. Quantile loss:

$$L_\tau(y, \hat{y}) = (y - \hat{y})(\tau - \mathbf{1}_{y<\hat{y}})$$

should be used as the primary forecast evaluation metric for stress prediction, following the BIS methodology.

9.5 Difference-in-Differences Design

The staggered adoption of SEBI's retail algorithmic trading framework across brokers between October 2025 and April 2026 provides a quasi-experimental variation. The proposed DiD model is:

$$Y_{b,t} = \phi_b + \psi_t + \rho \cdot \text{Adopt}_{b,t} + \xi W_{b,t} + \eta_{b,t}$$

where $Y_{b,t}$ is an outcome variable for broker b at time t (e.g., bid-ask spread, order imbalance, short-horizon autocorrelation); $\text{Adopt}_{b,t}$ is an indicator equal to one after broker b goes live with the algo framework; and $W_{b,t}$ contains broker-level controls. The identifying assumption is parallel pre-trends across early and late adopters.

9.6 SHAP Decomposition

For all tree-based models, global SHAP importance should be computed as:

$$\bar{\phi}_j = \frac{1}{N} \sum_{i=1}^{N} \mid \phi_j^{(i)} \mid$$

where $\phi_j^{(i)}$ is the Shapley value of feature j for observation i. SHAP dependence plots should be generated for the top ten features to identify the sign and shape of their marginal contribution to predictions, especially during stress episodes.

9.7 Robustness and Sensitivity Tests

Robustness dimension	Specific test
Sample stability	Rolling sub-period estimation (2010–2015, 2016–2020, 2021–2026)
Crisis partitioning	Separate estimation for 2013, 2018, 2020 stress episodes
Data quality	Winsorize at 1st and 99th percentiles; confirm with robust regressions
Model overfitting	Compare in-sample vs. out-of-sample performance gap
Transaction costs	Apply round-trip cost of 0.10–0.30% for portfolio backtests
Market tier	Separate estimation for large-, mid-, and small-cap universes
Alternative targets	Test on 1-, 3-, 6-, and 12-month return and volatility horizons
Alternative stress	Use alternative market-stress definitions (VIX-linked, spread-based)

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10. Expected Findings

10.1 Return Prediction

The most likely result is that tree-based ML models particularly XGBoost and Random Forest outperform CAPM and Fama–French in out-of-sample return prediction, with statistically lower RMSE and higher out-of-sample R^2. Deep learning architectures may add further incremental gains where temporal features are important, but the improvement over tree-based models is expected to be modest because monthly return prediction depends more on feature quality than architecture.

10.2 Volatility and Stress

LSTM and hybrid LSTM-XGBoost architectures are expected to perform best for volatility and stress forecasting, consistent with India-specific evidence. The relative advantage of ML over GARCH is expected to be largest at upper-tail quantiles and longer horizons, mirroring the BIS findings. Crisis subperiods should exhibit the largest ML advantage because stress dynamics are more nonlinear and the linear GARCH model cannot capture regime-switching behavior.

10.3 Portfolio Performance

AI-generated signals are expected to produce portfolios with Sharpe ratios 15–30 percent higher than classical-model portfolios before transaction costs. After costs, the advantage is expected to shrink but remain positive, especially for lower-turnover ML strategies based on monthly signals rather than daily ones.

10.4 Market Efficiency Implications

AI adoption is expected to reduce post-announcement drift after major corporate and policy events, consistent with faster information incorporation. However, the DiD analysis of retail algo adoption is expected to show a heterogeneous effect: broker segments that adopted early may show tighter spreads and lower autocorrelation, but small-cap markets may show increased volatility if algorithmic participation amplifies retail sentiment.

10.5 SHAP Interpretability

SHAP analysis is expected to identify momentum, liquidity, realized volatility, FPI flow intensity, and macro uncertainty (proxied by RBI communication sentiment or policy-rate volatility) as the dominant predictors of India-

specific market stress, paralleling the funding-liquidity and investor-overextension themes in the BIS stress literature.

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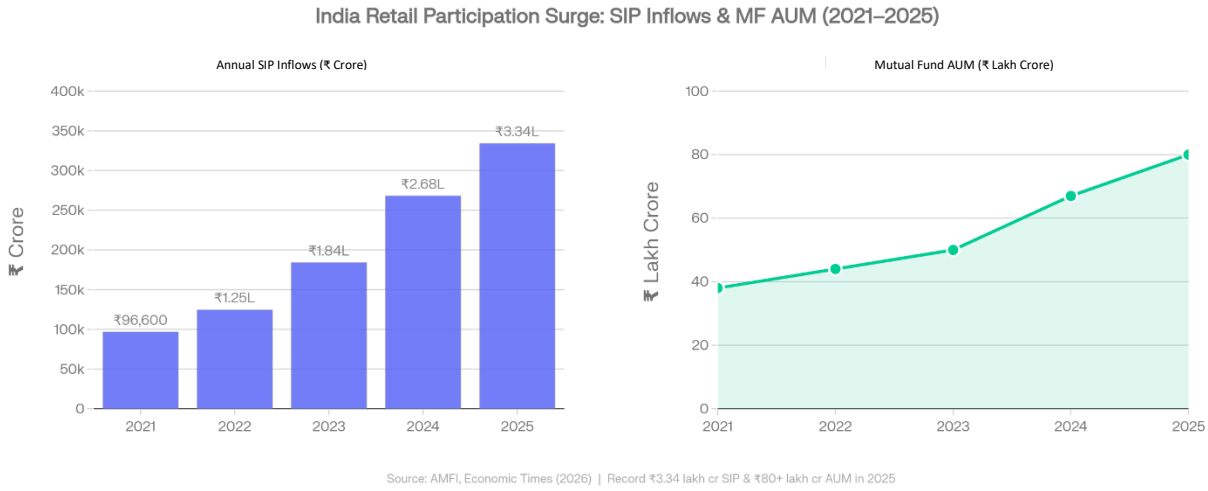
11. Policy and Practical Implications

11.1 For Investors and Portfolio Managers

AI-based models can improve security screening, tactical factor tilts, and stress-aware portfolio construction. The key recommendation is to treat ML signals as inputs to investment processes rather than as standalone allocation rules. Hybrid strategies that combine ML predictions with economic-discipline constraints such as factor exposure bounds or drawdown limits are likely to perform better than unconstrained black-box allocation.

***Figure 4*:** India SIP Inflows & MF AUM Growth (2021–2025)

SIP inflows surged from ₹96,600 crore in 2021 to a record ₹3.34 lakh crore in 2025, while mutual fund AUM more than doubled from ₹38 lakh crore to ₹80+ lakh crore, confirming the structural retail participation surge.



11.2 For Financial Institutions

Banks, brokers, and asset managers should invest in model-risk governance frameworks that include documentation, validation, backtesting, and continuous monitoring. The SEBI consultation's emphasis on explainability, bias detection, and periodic reporting should be treated as a business necessity rather than a compliance burden: institutions that can demonstrate interpretable and auditable AI will gain regulatory credibility and institutional client trust.

11.3 For SEBI and Securities Regulation

SEBI should consider using ML-based market surveillance tools to detect unusual trading patterns, liquidity fragmentation, and potential manipulation more effectively than rule-based systems. The retail algorithmic trading framework already creates an infrastructure of audit trails and registered strategies that is well suited to data-driven supervision. The empirical evidence from this paper's DiD design can help SEBI evaluate whether its framework is achieving the intended efficiency and investor-protection outcomes.

11.4 For the RBI and Macroprudential Policy

The RBI can use AI-based market-stress indices to improve the timeliness and granularity of financial-stability monitoring. The SHAP-based interpretability of ML stress models makes them usable for policy communication: when stress rises, SHAP can identify whether the dominant driver is currency pressure, liquidity withdrawal, earnings deterioration, or cross-asset contagion. This supports more targeted macroprudential responses rather than one-size-fits-all interventions.

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12. Limitations and Future Research

12.1 Limitations

The first limitation is data quality and availability. India's micro-level disclosure quality is improving but remains uneven across the size spectrum, and CMIE Prowess coverage of small-cap firms can be patchy. The second limitation is model instability: relationships that perform well in one subperiod may collapse when the macro or regulatory regime changes, and emerging markets are especially prone to structural breaks. The third limitation is that even with SHAP analysis, explainability is partial. Correlation-based attribution cannot establish causation, and feature importance can be unstable across bootstrap samples.

The fourth limitation is the retail algo DiD: if adoption is not truly exogenous but driven by broker characteristics correlated with market outcomes, the causal estimate may be biased. Careful pre-trend testing and robustness checks using alternative control groups will be essential.

12.2 Future Research Directions

Future research should extend this framework in five directions. First, intraday ML models for Indian equity markets can test whether algorithmic participation changes market microstructure quality in real time. Second, cross-asset AI models should examine spillovers between equity, currency, government bonds, and credit spreads using high-frequency Indian data. Third, generative AI and large language models should be examined for their effect on market narratives, investor sentiment, and information quality, consistent with IOSCO's emerging concerns about non-deterministic AI systems. Fourth, a comparative study of India, Brazil, Indonesia, and South Africa can identify which institutional moderators most strongly condition the value of AI in emerging markets. Fifth, the long-run evolution of market efficiency under increasing AI adoption deserves dedicated investigation, the question is whether AI drives markets toward efficiency asymptotically or generates new, technology-induced forms of instability.

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13. Conclusion

This paper has developed a comprehensive, policy-relevant, and academically rigorous empirical framework for evaluating whether financial machine learning and artificial intelligence improve asset pricing, market efficiency, portfolio choice, and market-stress prediction in India. The case for doing this study in India in 2026 is stronger than it has ever been: the country combines 7.6 percent GDP growth, a 25-crore investor base, record SIP inflows, a market-cap-to-GDP ratio above 130 percent, an active SEBI AI governance consultation, and a new retail algorithmic trading framework that creates natural empirical variation.

The paper argues that AI is not a generic performance-enhancing technology in finance. Its value is conditional: it is highest where financial relationships are nonlinear, where information is heterogeneous and dispersed, and where stress regimes create tail dynamics that linear models suppress. India satisfies these conditions strongly, and the institutional context including SEBI's regulatory engagement makes the policy translation of empirical results particularly direct.

The deeper implication is that the coevolution of AI capability and regulatory architecture in India is now rapid enough that academic research can no longer treat the two as separate. The value of an AI system in Indian capital markets depends not only on its technical performance but on whether it can be governed, explained, and trusted by the participants and supervisors that determine its scale of adoption. That joint standard comprising technical performance plus institutional accountability is the right benchmark for financial AI in emerging economies.

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Appendix A: Variable Definitions and Construction

Variable	Definition	Construction	Frequency	Source
Excess return	Adjusted return less risk-free rate	Monthly log return less 91-day T-bill rate	Monthly	NSE, RBI
Realized volatility	Daily squared returns aggregated monthly	$\sum_{d=1}^{D} r_d^2$	Monthly	NSE
Amihud illiquidity	Price impact per unit of trading volume	$\| r_d \| / \text{Volume}_d$, monthly avg	Monthly	NSE
Market stress index	Composite of vol., correlation, liquidity, FPI	Normalized z-score composite	Daily/Weekly	NSE, RBI, Bloomberg
AI signal	Model output score	Predicted return or stress probability	Daily/Monthly	Estimated
Sentiment score	Textual tone from news	NLP embedding, avg. polarity by firm-day	Daily	News feeds
Size (log ME)	Log of market capitalization	Log(Price × Shares outstanding)	Monthly	CMIE Prowess
Value (B/M)	Book equity to market equity	Book equity / Market cap	Quarterly	CMIE Prowess
Profitability (RMW)	Operating profitability	Operating income / Book equity	Annual	CMIE Prowess
Investment (CMA)	Asset growth	Year-on-year change in total assets	Annual	CMIE Prowess
GDP growth	Real GDP growth	Quarter-on-quarter seasonally adjusted	Quarterly	MOSPI
Inflation	CPI inflation	Year-on-year CPI change	Monthly	MOSPI, RBI
Repo rate	RBI benchmark rate	Point-in-time policy rate	Event-based	RBI
Global risk	Market uncertainty	CBOE VIX, daily close	Daily	Bloomberg

Appendix B: Model Architecture Summary

Model	Class	Hyperparameters	Validation	Key strengths	Key limitations
CAPM	Traditional	—	—	Interpretability, theoretical discipline	Single-factor, linear only
Fama–French 3F/5F	Traditional	—	—	Multi-factor risk decomposition	Linear, cannot capture interactions
ARIMA(p,d,q)	Time series	p, d, q via AIC/BIC	AIC	Parsimonious mean dynamics	No volatility clustering
GARCH(1,1)	Time series	ω, α, β via MLE	Log-likelihood	Volatility persistence	Linear conditional variance
Ridge/Elastic Net	Classical ML	λ, l1 ratio via cross-val	K-fold	Regularized linear; handles multicollinearity	Still linear in features
SVM	Classical ML	C, γ, kernel	K-fold	Nonlinear boundary; good for classification	Slow on large samples
Random Forest	Tree-based ML	n_trees, max_depth, mtry	OOB error	Nonlinear, interpretable via SHAP	Can overfit with many features
XGBoost	Tree-based ML	depth, η, λ, subsample	Walk-forward	Best tabular ML; fast; SHAP-compatible	Hyperparameter-sensitive
Feedforward NN	Neural network	Layers, units, dropout	Walk-forward	Flexible approximation	Black-box; overfitting risk
LSTM	Sequence model	Units, dropout, seq. len.	Walk-forward	Temporal memory; volatility patterns	Data-hungry; complex
Transformer	Attention model	Heads, depth, d_model	Walk-forward	Long-range attention	Very data-hungry; slow
Hybrid LSTM-XGB	Ensemble	Combined	Walk-forward	Best of temporal + tabular features	High complexity

Appendix C: Hypothesis Mapping Summary

Hypothesis	Prediction	Test	Benchmark	Expected sign
H1	ML > classical in return RMSE	Diebold-Mariano test	CAPM, FF5	Negative ΔRMSE for ML
H2	ML advantage ↑ in stress	Regime-interaction regression	GARCH	Positive interaction with stress indicator
H3	ML Sharpe > classical Sharpe	Portfolio backtest	Equal-weight benchmark	Positive ΔSharpe net of costs
H4	AI reduces post-event drift	Event study + VAR	ARIMA	Reduced abnormal return autocorrelation
H5	Sentiment adds predictive power	Feature ablation test	Financial variables only	Positive ΔAUC with sentiment
H6	SHAP identifies liquidity/uncertainty	SHAP global importance	—	Liquidity, vol, macro in top-5 features
H7	Algo adoption ↓ bid-ask spread	DiD: early vs. late adopters	Pre-adoption baseline	Negative coefficient on Adopt indicator